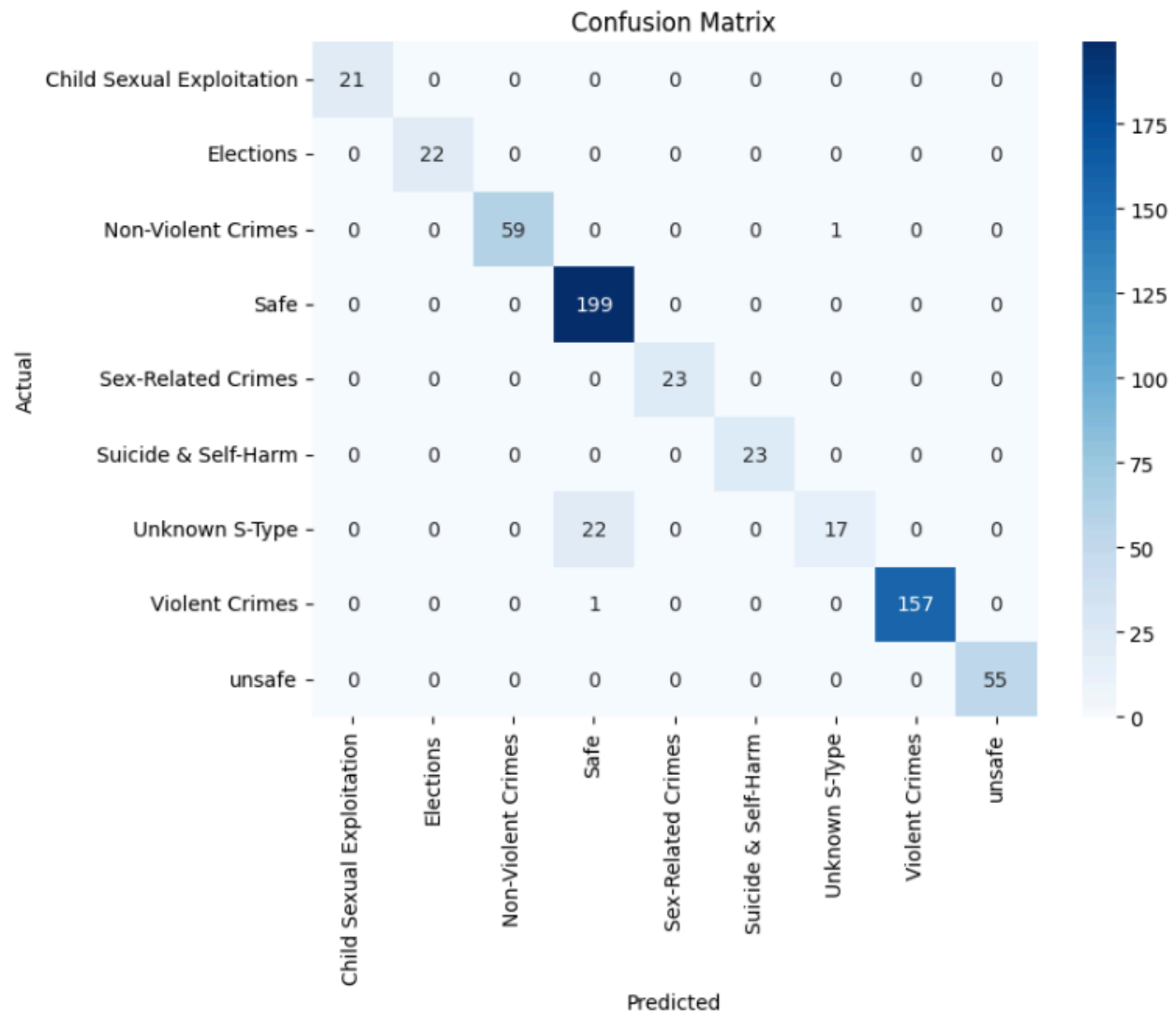
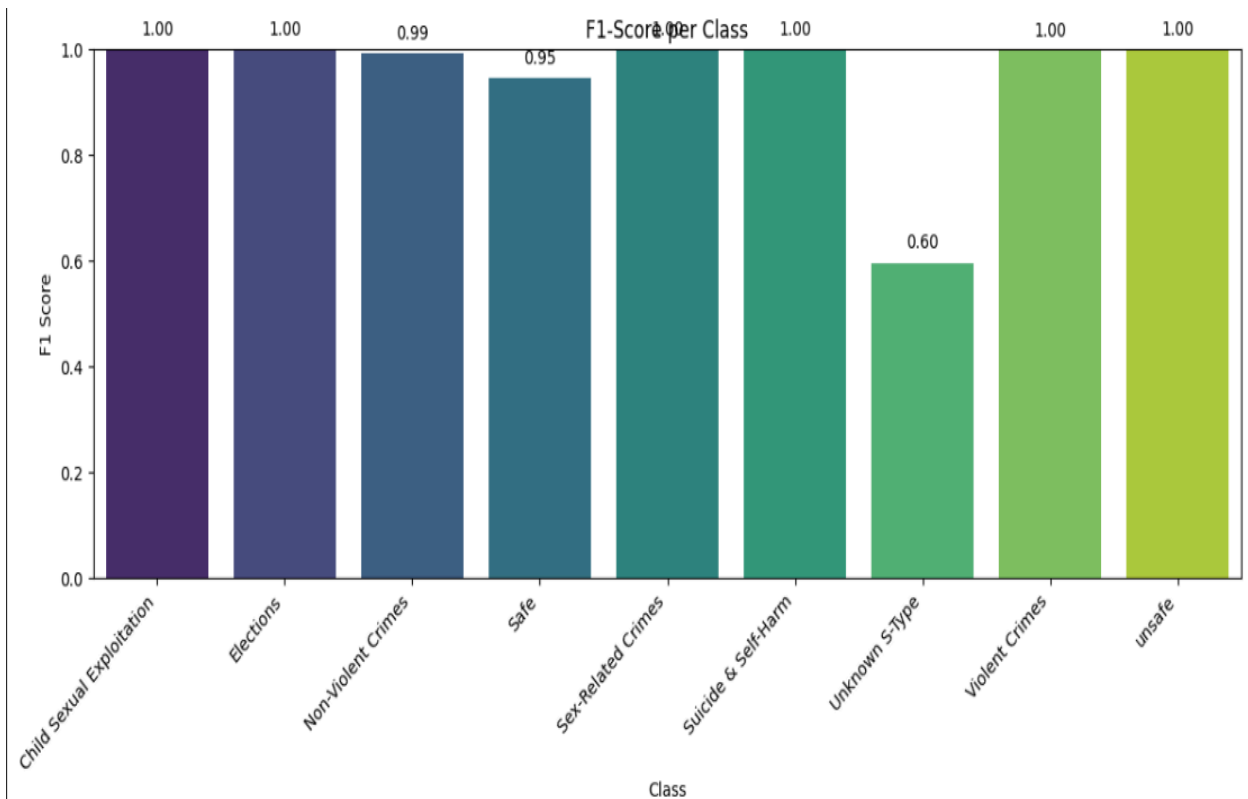


Results:



- only class 6 shows a recall problem which drags down the macro average.
- Classes 0,1,2,3,4,5,7,8 → high precision and recall
- Class 6 → precision = 0.94 and the recall = 0.44 and the F1 = 0.60 → clearly the bottleneck



- most classes have $F1 \approx 1 \rightarrow$ excellent performance
- class 6 $F1 = 0.60 \rightarrow$ main area for improvement

LoRA's benefits:

1. faster training
2. memory efficiency
3. preserved knowledge
4. flexible adaptation to new tasks